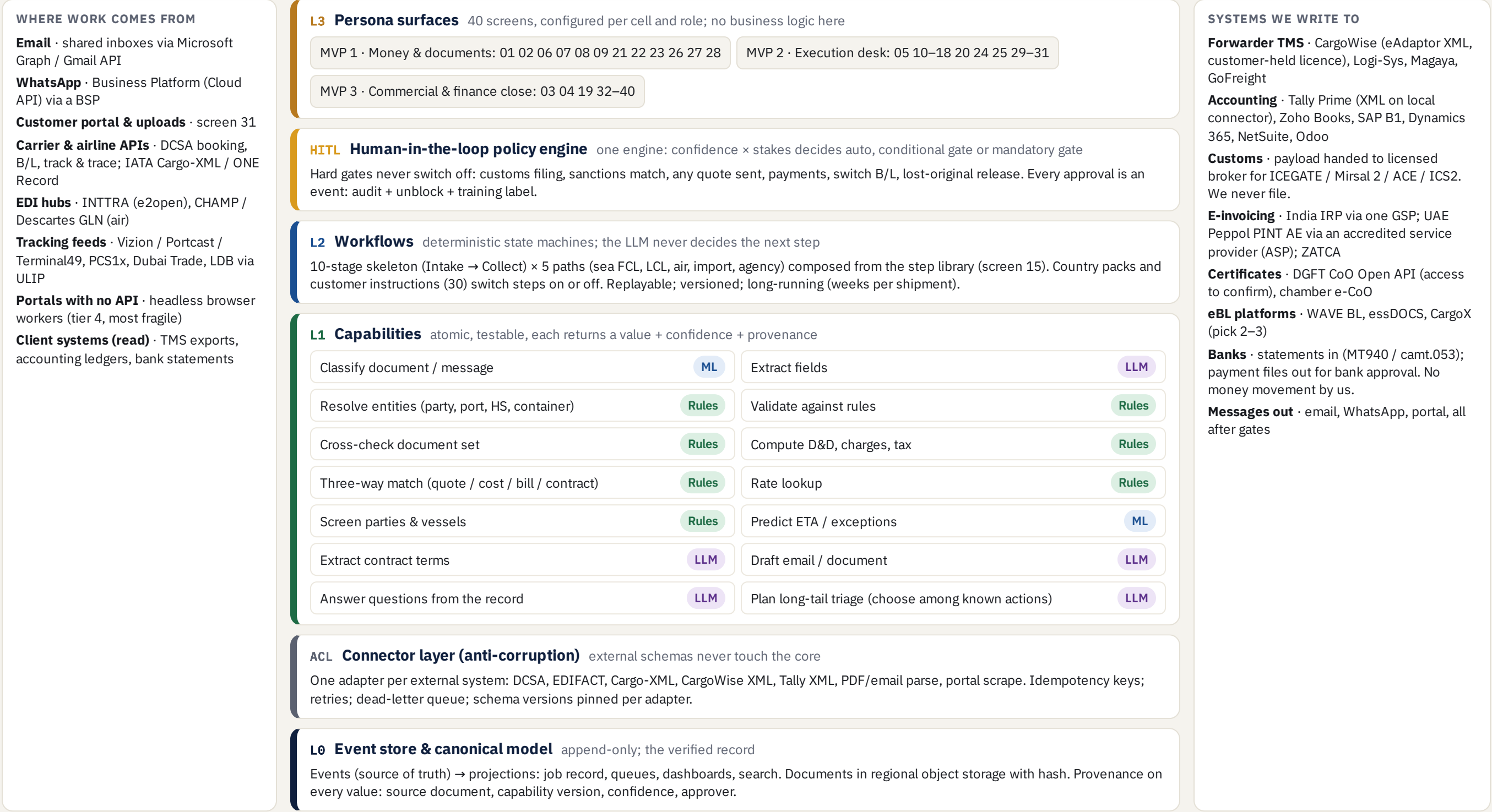


System architecture: how the 40 screens are powered

Four layers from the Platform Systems Architecture, extended to the full product. Intelligence lives in capabilities, control in deterministic workflows, truth in the event log. Everything outside sits behind an adapter.



Integration map: every system we connect to, and when

Grouped by counterpart. "First needed" is the MVP that cannot ship without it. Facts checked against live sources in Sep 2026; items marked "to confirm" are open with the provider.

COUNTERPART	WHAT WE EXCHANGE	DIRECTION	METHOD	FIRST NEEDED	SCREENS	RISK / DECISION
CHANNELS						
Email (M365 / Google Workspace)	All inbound enquiries, documents, vendor bills; outbound drafts	In / out	Microsoft Graph with webhooks; Gmail API with Pub/Sub push	MVP 1	01 09 27	Tenant admin consent; Gmail restricted scopes need a third-party security assessment (CASA)
WhatsApp	Customer updates, document requests, collection nudges, trucker chat	In / out	WhatsApp Business Platform (Cloud API) via a BSP	MVP 2	09 31	Per-message pricing; Meta "AI Providers" policy (Feb 2026) to review before launch
Customer portal / upload	Documents, status, invoices, instructions	In / out	Own web app + signed upload links	MVP 2	31	—
FORWARDER SYSTEMS OF RECORD						
CargoWise	Jobs, parties, documents, charges, invoices	Read MVP 1 · write MVP 2	eAdaptor (Universal XML over HTTPS), enabled on the customer's licence	MVP 1	05 12 21 34	WiseTech controls partner access and is a potential competitor. Stay email/document-first so we never depend on it.
Logi-Sys (Softlink) and other TMS	Same as above	Read → write	API if available; otherwise database views or scheduled exports + import files	MVP 1	05 12 21	API unverified; expect file-level integration first
Tally Prime	Post sales invoices, purchase bills, receipts; read ledgers	Write / read	XML over HTTP (port 9000) via a small on-premise connector	MVP 1	21 22 27 34	Runs on the client's machine; connector must survive restarts and offline periods
Zoho Books, SAP B1, Dynamics 365, NetSuite, Odoo	Same as Tally	Write / read	Vendor REST / OData APIs	MVP 1	21 22 27 34	Build one accounting interface, one adapter each
CARRIERS, AIRLINES, VISIBILITY						
Ocean carriers (direct)	Booking, SI, draft B/L, tracking events, D&D charges, schedules	Out / in	DCSA Booking 2.0, B/L 3.0, T&T 2.2 APIs where offered (Maersk, Hapag-Lloyd, CMA CGM...)	MVP 2	05 10 11 25	Uneven coverage; T&T 3.0 still beta
INTTRA (e2open)	Multi-carrier booking, SI, VGM, status	Out / in	EDIFACT / XML / API	MVP 2	05 10 11	Integration package fees; long-tail reach
Airlines & hubs	Bookings, e-AWB (FWB/FHL / XFWB), status (FSU)	Out / in	Cargo-XML / Cargo-IMP via CHAMP or Descartes; WebCargo / cargo.one for rates	MVP 2	16	ONE Record is preferred, not mandatory: build Cargo-XML now
Tracking aggregator	Container milestones, predicted ETA, D&D risk	In	Vizion / Portcast (API + webhooks); India inland via LDB through ULIP	MVP 2	25 18	Per-container fees; pick one, keep carrier direct as backup
Port community systems	e-DO, container status, CFS/ICD events, gate passes	In / out	PCS1x (India) APIs, Dubai Trade, Maqta; else portal automation	MVP 2	18 17	UAE API scope unverified
Carrier & port portals (no API)	D&D statements, DO status, invoices	In	Headless browser workers behind an adapter	MVP 1	27 18	Most fragile path; monitor heavily; respect site terms
CUSTOMS, ORIGIN, COMPLIANCE						
Customs broker (licensed)	Declaration payloads, status, queries	Out / in	Structured payload + documents to the broker's system; status back by email or API	MVP 2	11 18 38	We prepare and validate; the broker files (liability boundary)
ICEGATE (India)	Status reads; payload format	In	JSON Open APIs (e-seal / DSC) or via broker; ULIP for status	MVP 2	11 18 38	Third-party software eligibility to confirm
Mirsal 2 / Dubai Trade, ATLP	Declaration status, DO, permits	In	Via broker; APIs for registered parties	MVP 2	18 38	API details not public
DGFT CoO e-platform	Certificate of origin application and status	Out / in	CoO Open API (launched Sep 2026; signed payloads, IP whitelisting)	MVP 3	07 38	Access for third-party SaaS to confirm
Sanctions lists	Party and vessel screening	In	OFAC SLS, US CSL API, UN, EU FSF, UK Sanctions List, UAE EO CN, India MHA; or a licensed vendor	MVP 2	24	Embedding a commercial list needs an OEM / reseller licence
MONEY, TAX, DOCUMENTS OF TITLE						
E-invoicing: India	IRN + signed QR on sales invoices and credit notes; e-way bill; GSTR-2B / IMS for input credit	Out / in	One GSP (REST)	MVP 3	21 36	Forwarders above ₹5 cr turnover are in scope. Until MVP 3, invoices go into the client's accounting system, which generates the IRN
E-invoicing: UAE	Structured invoices out and in	Out / in	Peppol PINT AE via an accredited service provider (ASP)	MVP 3	21 27 36	Mandatory 1 Jan 2027 for revenue ≥ AED 50m. Until MVP 3 we post into the client's accounting + ASP; later decide: become, partner or plug in
Banks	Statements for cash application; payment files for approval	In / out	MT940 / camt.053 via host-to-host SFTP or bank APIs; UAE Open Finance and India Account Aggregator need a licensed partner	MVP 3	22 35	We never move money; the bank's own approval stays
eBL platforms	Receive, endorse, surrender electronic B/Ls	In / out	WAVE BL / essDOCS / CargoX APIs; DCSA PINT when mature	MVP 3	10 18	India Bills of Lading Act 2025 recognises eBLs; UAE onshore status unverified

Ingestion tiers, as in the platform architecture: documents and messages are the main road for this segment; APIs are the accelerant where a client has them; EDI hubs cover the carrier long tail; portal automation is last resort. Every row sits behind its own adapter (A1, connector layer). Full source list with URLs, licences and verification status: *Integration & data catalogue* workbook.

Reference data and external databases

What each dataset powers, where it comes from, and what it costs. The deterministic checks behind 08, 21, 24, 27 and 38 depend on these tables, not on the model.

Dataset	Drives which check (screen)	Primary / Official source	Commercial option	Refresh	Licence note	MVP
Classification & Duty						
HS nomenclature (6-digit)	HS on invoice vs packing list vs declaration (08, 38)	WCO (licensed database); free public HS lists for codes only	—	5-yearly; HS 2028 on 1 Jan 2028	Description text is WCO copyright	MVP 1
India tariff (8-digit) + duty notifications	Bill of entry prep, duty estimate (18, 38)	CBIC / ICEGATE (PDF, web)	Taxmann, Cen-Cus parsed data	Budget + ad-hoc notifications	No official tariff API found	MVP 2
UAE / GCC tariff (12-digit)	Mirsal 2 payload, CEPA duty (18, 38)	Dubai Customs integrated tariff	—	Ad-hoc; 12-digit phased to Feb 2027	Model must store 12 digits	MVP 2
US HTS, EU TARIC, UK tariff	Duty in quotes to US / EU / UK (03)	USITC, EU TARIC, UK Trade Tariff API (key needed)	Descartes CustomsInfo, Avalara, ONESOURCE	Daily to weekly; US overlays change weekly	Free	MVP 3
Classification rulings	Evidence for HS suggestions (38)	CBP CROSS, EU EBTI, UK ATaR, India CAAR	—	Continuous	Web only; retrieval index, not training	MVP 3
Codes & Identities						
UN/LOCODE + Indian customs codes	Port and place normalisation everywhere	UNECE (free CSV); CBIC location codes	—	Twice a year	Free	MVP 1
ISO 6346 / BIC prefixes	Container number check digit and owner (08, 10, 11)	Algorithm free; BIC register	BIC BoxTech API	Continuous	Owner lookup is paid	MVP 1
IATA airport, airline, AWB prefix	AWB validation, air routing (16)	IATA code files (licensed)	OurAirports as a free fallback	Monthly	Paid licence	MVP 2
SCAC, IMO vessel numbers	US filings; vessel identity and sanctions (24)	NMFTA (paid); IMO GISIS (free registration)	Kpler / Lloyd's List for vessel risk	Continuous	—	MVP 2
GSTIN, PAN, IEC, company registries, LEI	Customer and vendor onboarding; invoice validation (24, 35)	GSTN via GSP, DGFT IEC, MCA via aggregators, GLEIF (free)	Signzy, Probe42, Perfios	Real time	Paid via vendors	MVP 2
Compliance						
Sanctions & denied parties	Screen every party on every job (24)	OFAC, US CSL, UN, EU FSF, UK Sanctions List, UAE EOCN, India MHA + DGFT denied entities	OpenSanctions, World-Check, Dow Jones, Descartes Visual Compliance	Daily	OEM licence to embed commercial lists	MVP 2
Dual-use / SCOMET / export control	Commodity flags at booking (05, 24)	BIS EAR (eCFR), EU Annex I, DGFT SCOMET list	—	Ad-hoc	Free	MVP 3
Dangerous goods	DG checks at booking and SI (05, 10)	UN numbers list (free)	IMDG Code (IMO), IATA DGR (licensed)	Biennial / annual	Licensed; no verbatim display	MVP 2
Commercial & Money						
Carrier D&D tariffs	Vendor bill audit and customer rebill (27, 21)	Carrier websites and PDFs per port	Maersk / Hapag D&D APIs where offered	Ad-hoc; snapshot with dates	No standard feed: we maintain tables	MVP 1
Contracts & terms	Free time, rates, dispute windows, liability (28)	Client contracts + amendment emails (extracted, confirmed by a person)	—	On change	Client data	MVP 1
Customer instructions (SOPs)	Account rules at each step (30)	Client SOPs, emails, contracts	—	On change	Client data	MVP 2
Rate sheets & surcharges	Quotes, surcharge checks on vendor bills (03, 32, 27)	Carrier rate sheets (Excel by email), advisories	Freightify, carrier spot APIs; FBX / Drewry WCI / Xeneta as benchmarks	Weekly / ad-hoc	Benchmarks need licence to redistribute	MVP 3
FX rates	Customs value (CBIC rate), billing and close (FBIL, CBUAE, ECB)	CBIC notifications, FBIL, CBUAE, ECB API	OANDA, Open Exchange Rates	Daily; customs fortnightly	Free	MVP 1
Tax & e-invoice rules	GST split, VAT, withholding, e-invoice schema (21, 36)	Country packs from GSTN, UAE FTA / MoF, ZATCA	GSP / ASP rule engines	On notification	Validate per country with advisers	MVP 1
LC rules	LC document checks (08)	ICC UCP 600, ISBP 821, eUCP (licensed)	—	Rare	Encode checks; do not show ICC text	MVP 1
Emissions factors	Customer emissions reports (40)	GLEC Framework v3.2 (free), ISO 14083	EcoTransIT, Searoutes, carrier APIs	Periodic	—	MVP 3

HOW REFERENCE DATA IS HELD

Every table is versioned with **valid-from / valid-to** and **as-of**. A check records which version it used, so a disputed D&D invoice or a customs query can be replayed exactly as of that day.

BUILD VS BUY

Buy or download: codes, tariffs, sanctions lists, FX. Build and own: D&D tariff tables, contract terms, customer instructions, validation rules. The owned tables are the moat; the model is not.

TEST DATA FOR THESE CHECKS

The synthetic data pack includes carrier free-time agreements, an email amendment (v2 from 12 Jul) and container events, so the D&D rules can be tested against the correct contract version.

Agents mapped to capabilities, data and gates

The twenty agents in the roster (26), decomposed into the shared capabilities of layer 1. Tiers show where each agent sits on the determinism ladder: rules for money and control, models only at the language edge.

AGENT	WHAT IT DOES, AS CAPABILITIES	TIERS	READS	WRITES (EVENTS)	EXTERNAL SYSTEMS	HUMAN GATE	SCREENS
Enquiry	Classify, extract, resolve, draft	ML LLM Rules	Email, WhatsApp, customer master	ShipmentInitiated, FieldsExtracted, DraftCreated	Graph / Gmail, WhatsApp	Commercial terms	01 02
Routing order	Extract, validate vs agency agreement	LLM Rules	Partner emails, agency terms (28)	RoutingOrderRead, ShipperBriefed	Email	Deviating from partner	19
Rate & quote	Rate lookup, compute margin, draft	Rules LLM	Rate sheets (32), contracts, benchmarks	RatesRetrieved, QuoteGenerated	Carrier spot APIs, WebCargo	Every quote sent	03 04
Booking	Validate prerequisites, map to carrier format	Rules	Job record, cut-offs, DG and sanctions results	BookingCreated, BookingConfirmed	DCSA / INTTRA / airline hubs, TMS	Booking submit (configurable)	05
Document	Classify, extract, cross-check, draft SI / HBL / HAWB	ML LLM Rules	All documents, document rules (13), LC	DocumentClassified, FieldsExtracted, ExceptionRaised	DCSA B/L, Cargo-XML	Low-confidence fields	07 08 10
Messages	Classify intent, file to job, draft, chase rules	ML LLM Rules	Mailboxes, job record, customer instructions (30)	MessageFiled, ReplyDrafted	Graph / Gmail, WhatsApp	Client emails (configurable)	09
Milestone	Normalise events, predict ETA	Rules ML	Tracking feeds, schedules	MilestoneRecorded, ETACHanged	Carrier T&T, aggregator, PCS	—	11 16 17 25
Exception	Rank by impact, propose next action	Rules ML LLM	All open jobs, milestones, cut-offs	ExceptionRaised, ActionProposed	—	Any resolution	06 25
Customs prep	Resolve HS, build payload, validate	Rules LLM	Invoice, packing list, tariff, rulings, FX	CustomsDeclarationPrepared, HandedToBroker	Broker channel, ICEGATE / Mirsal status	Any filing	11 18 38
KYC & screening	Screen parties and vessels, onboarding checks	Rules ML	Sanctions lists, registries, GSTIN / IEC	SanctionsScreeningRun, KYCStatusChanged	Sanctions feeds, KYC vendors	Any match	24
Compliance	Deadlines, licences, access monitoring	Rules	Country packs, event log	ComplianceTaskRaised	—	Filing and licence decisions	24
Delivery / POD	Read e-POD, log deviations, fire billing trigger	ML Rules	POD images, trucker messages	PODRecorded, DeviationLogged	Trucker apps, WhatsApp	Claims on deviations	20
Billing	Build invoice from job record, tax per pack	Rules	Job record, quote, customer agreement	InvoiceDrafted, InvoiceIssued	TMS / accounting, IRP (GSP), UAE ASP	Variance over tolerance	21
Revenue assurance	Three-way match, find events with no charge	Rules	Quote, costs, vendor bills, events	LeakageFound	—	Match exceptions	21 23
Vendor bill audit	Extract bill, check events, rates, contract version, D&D maths	LLM Rules	Vendor bills, container events, contracts, D&D tariffs	BillAudited, DisputeDrafted, RebillProposed	Email, portals, accounting	Pay, dispute or rebill (one click)	27
Contract	Extract and version terms from documents and emails	LLM Rules	Contracts, amendment emails	TermsExtracted, TermsConfirmed	Email, document store	Putting terms live	28
Credit note	Check reason codes, route by threshold	Rules	Invoices, disputes	CreditNoteRequested	Accounting, IRP / ASP	Over threshold	22
Collections	Ageing ladder, cash application	Rules ML	Invoices, bank statements	ReminderSent, PaymentMatched	Banks (MT940 / camt.053), accounting	Credit hold, 61+ escalation	22
Dispute	Log, assign, root-cause	LLM Rules	Emails, invoices, job record	DisputeLogged, DisputeResolved	Email	Any outcome	22
MIS	Answer questions from the record and BI models	LLM Rules	Projections, BI models	AnswerGiven	BI tool	Sharing answers	23

AN AGENT IS CONFIGURATION, NOT A SERVICE

Each agent = a bundle of capabilities + the workflow steps it may act in + an autonomy level (26) + a gate. Twenty agents share about fourteen capabilities: build the capabilities once.

EVENT NAMES

Extend the catalogue in the Canonical Data Model with the new families shown here (billing, audit, collections, contracts). Events are immutable: add types, never change meaning.

TEST EACH CAPABILITY, NOT EACH AGENT

Golden sets are per capability (extraction per document type, D&D maths, three-way match, intent). The synthetic pack's labels map to these directly.

Deployment, security, evaluation and build order

How the platform is hosted and trusted, how quality is measured before anything ships, and which platform pieces each MVP pays for.

DEPLOYMENT & DATA RESIDENCY

Control plane (tenants, config, releases) separate from **regional data planes**.

India plane: AWS Mumbai / Hyderabad or Azure India.
UAE plane: AWS me-central-1 or Azure UAE North. Add planes per country as clients need.

Partition key decided before the first event (per tenant is simplest).

LLM and document-AI inference in the same region as the data, or with PII redacted first. Check model availability per region.

Backups in-region; restore tested quarterly.

Rules to meet: India DPDP (most duties from ~mid-2027), CERT-In 6-hour incident reporting and 180-day Indian logs, UAE PDPL, DIFC / ADGM if clients sit there.

SECURITY & TRUST

SSO (Entra ID / Google) + role-based access by cell and role (14); least privilege per agent.

Tenant isolation at the data layer; row-level policies; per-tenant encryption keys.

HSM for digital signatures (DSC / e-seal for ICEGATE, DGFT); secrets in a vault.

Immutable audit: every action is an event with who, what, source and confidence.

Email and WhatsApp access scoped to named shared mailboxes only.

Right to erasure in an append-only log: crypto-shredding of personal fields.

Roadmap: ISO 27001 then SOC 2 Type II. Clients will ask in diligence.

DATA, EVALUATION & LEARNING LOOP

Golden sets per capability: extraction by document type, cross-checks, D&D maths, intent, contract terms.

Three sources: the synthetic pack (labelled, 24 error types) → 200–500 redacted client documents per type under NDA → corrections captured in production.

Release gate: no capability version ships if its golden-set score drops.

Drift watch: confidence and override rates per capability, per client, per document issuer.

Flywheel: every human correction is a labelled event; move tasks down the ladder (LLM → small model → rule) as data accumulates.

PROPOSED STACK (TO VALIDATE WITH THE CTO)

Event store & projections: Postgres (append-only events + outbox), OpenSearch for search.

Workflows: Temporal (deterministic, replayable, long-running).

Messaging: Kafka / Redpanda or cloud queues for event fan-out.

Documents: regional object storage with hashes and KMS.

Document AI: OCR / layout service + LLM vision behind one LLM gateway.

Rules: rules-as-code with effective dates; country packs as config.

Retrieval: pgvector over contracts, SOPs and rulings.

Portal workers: Playwright, isolated per tenant.

Observability: OpenTelemetry; adapter health board.

Front end: React; live updates over websockets.

MVP	SCOPE	PLATFORM PIECES BUILT	INTEGRATIONS SWITCHED ON	DEPLOY PER CLIENT
MVP 1	Money & documents	Email ingestion, document AI, event store, rules engine, review queue, contracts repository, D&D tariff tables, vendor bill audit, three-way match, accounting write (Tally / Zoho / SAP B1), TMS read (exports or eAdaptor), FX feeds	Microsoft Graph / Gmail, accounting APIs, CargoWise read, carrier portals for D&D statements	2–4 weeks per client; no system replaced
MVP 2	Execution desk	Workflow engine across 5 paths, TMS write-back, carrier and airline integrations, tracking aggregator, customs payload builder, sanctions screening, WhatsApp, customer portal	DCSA / INTTRA, Cargo-XML hubs, Vizion / Portcast, PCS1x, broker channel, sanctions feeds	6–10 weeks per client
MVP 3	Commercial & finance close	Rate ingestion and quoting, e-invoicing, tax reconciliation, payments files, partner settlement, claims, emissions, eBL	GSP (India), UAE ASP / Peppol, ZATCA, banks, eBL platforms, DGFT CoO API, rate benchmarks	Module by module

DECISIONS TO TAKE NOW

- Hosting regions and data-plane partition key (before the first event).
- CargoWise stance: read via client-enabled eAdaptor, never a dependency for MVP 1.
- Sanctions: government lists in-house vs licensed vendor with OEM rights.
- UAE e-invoicing: become an ASP, partner with one, or plug into each client's ASP (large clients live 1 Jan 2027).
- India: one GSP; apply for ULIP; confirm ICEGATE and DGFT API eligibility; buy an HSM.
- Which 2–3 eBL platforms first.
- Licensed data budget: HS database, IATA codes, IMDG / DGR, ICC rules.